

The 2024 Programming Language Landscape at a Glance

State of Developer Ecosystem 2024 (JetBrains, n=23,262 developers)

JavaScript

61%

Most used overall (down from 70% peak in 2020)

Python

57%

Rapid rise from 32% in 2017; #1 primary language at 35%

TypeScript

37%

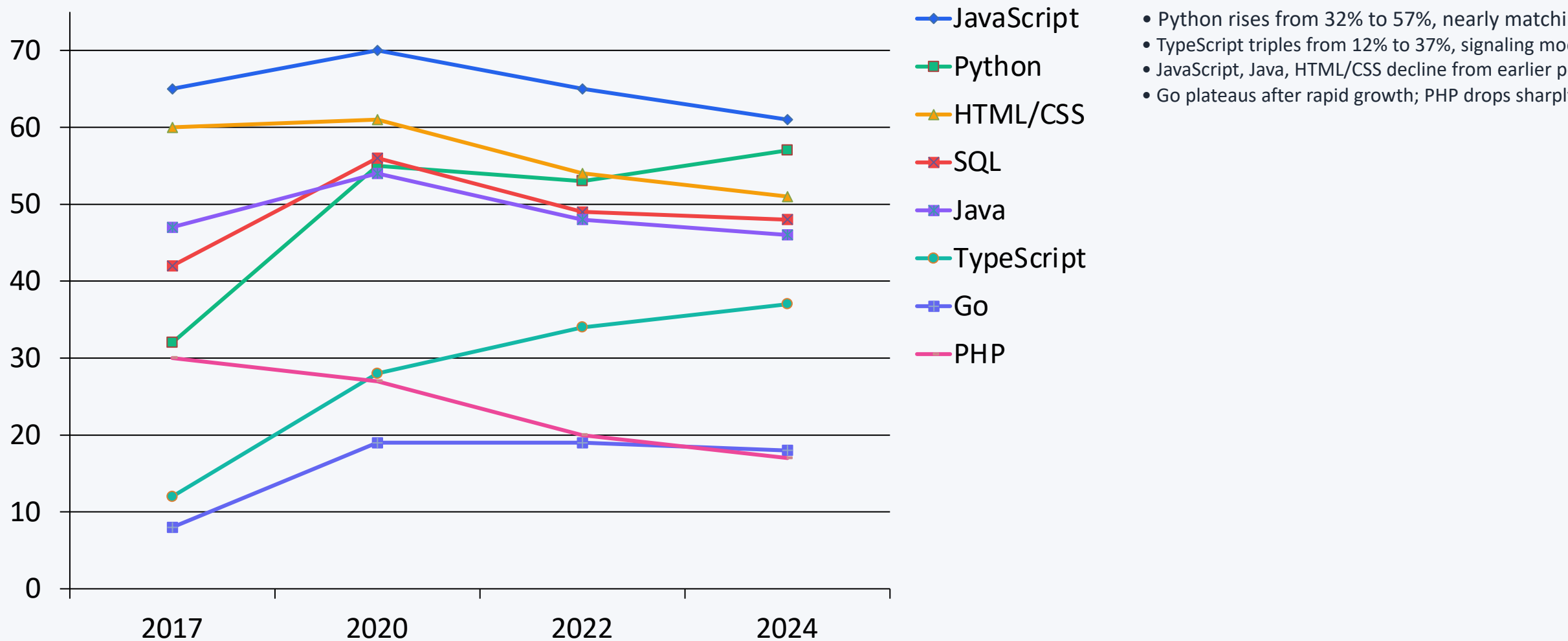
Tripled from 12% in 2017; top satisfaction score 199

Go + Rust

18% / 11%

Leading adoption momentum for infra and systems work

Language Momentum: 8 Years of Usage Shifts (2017-2024)



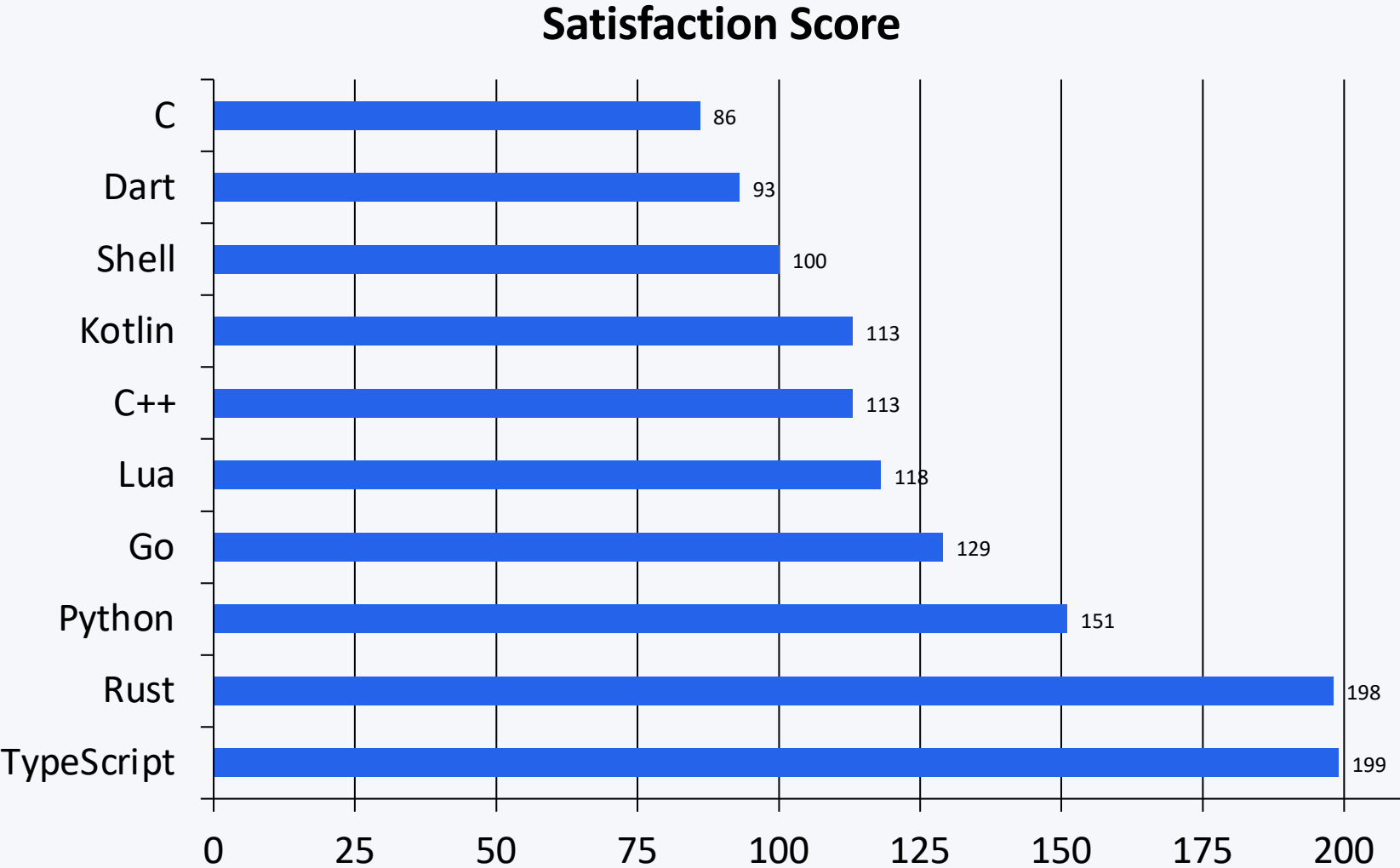
Primary Language vs Adoption Intent: Developers Prefer Stability

Language	Primary Language	Plan to Adopt
Python	35%	6%
Java	30%	4%
JavaScript	28%	3%
TypeScript	19%	5%
SQL	17%	4%

Key Interpretation

- Adoption intent stays below 6% for all major languages
- Developers are heavily invested in existing ecosystems
- Hiring should emphasize deep ecosystem expertise

Most Loved Languages in 2024: What Developers Actually Enjoy Using



Why this matters

- TypeScript and Rust lead by a wide margin.
- Top languages cluster around type safety and modern tooling.
- Satisfaction signals retention and hiring attractiveness.

Language-Domain Fit: The Right Tool for the Right Job

Application Domain	Leading Language	Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

What This Means for Your Technology Strategy in 2025

1) Prioritize Python for AI/ML capacity

Python is dominant in AI/ML (31%), data science (28%), and remains near-top in overall usage.

2) Standardize on TypeScript for frontend/full-stack

TypeScript leads UI development (62%) and ranks #1 in satisfaction (199).

3) Build Rust/Go paths for platform teams

Go (18%) and Rust (11%) are the strongest adoption plays for reliability-focused infra.

4) De-risk legacy stack hiring

PHP drops from 30% to 17% and Ruby from 10% to 4%, indicating shrinking talent momentum.

5) Treat developer experience as strategy

Higher satisfaction aligns with safety/tooling quality; use this in retention and platform investment decisions.